Eduwide App

Educational Management System — Full Project Report

PLATFORM
Flutter (Multi-Platform)

BACKEND
Firebase (Firestore + Auth)

VERSION
1.0.0+1

DEVELOPER
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INSTITUTION
Sri Lanka Advanced Technological Instatute Galle- Higher National Diploma In Information Technology (HNDIT4232 Mobile Application Development Assignment)

SDK
Dart ≥ 3.9.0 < 4.0.0

Android | iOS | Web | Windows | Linux | macOS | July 2026

1. Project Overview

Eduwide is a cross-platform mobile/desktop application built with Flutter, designed as a comprehensive **Educational Management System and Login Portal**. The app provides a powerful set of modern features designed to digitize the learning experience. These include **Zoom Class integration** for teachers to create, schedule, and launch live video sessions directly from the platform. It features robust **Add Students functionality** for seamlessly enrolling students into specific classes and tracking their academic journey based on streams and subjects. Additionally, the system includes a complete **Payment Gateway Option**, allowing administrators and teachers to monitor pending and completed payments effortlessly. By bringing these tools together under one unified portal, Eduwide simplifies administrative and educational tasks for Admins, Teachers, and Students alike.

Key Goals

- Digitize and streamline live learning via Zoom meeting integration
- Allow teachers and admins to add, manage, and enroll students into classes efficiently
- Monitor class fees through an integrated payment tracking option (Pending/Completed payments)
- Enable role-based access for Administrators, Teachers, and Students
- Secure user authentication, extensive profile management (NIC, Stream, Subjects), and status tracking
- Allow admins to upload, manage, and distribute learning resources universally

2. Technology Stack & Dependencies

Category	Package	Version	Purpose
UI Framework	flutter	SDK	Core UI
Database	cloud_firestore	^6.0.0	NoSQL real-time database
Authentication	firebase_auth	^6.0.0	User login/session
Core Firebase	firebase_core	^4.0.0	Firebase initialization
Timezone	timezone	^0.10.1	Asia/Colombo timezone support
Date Formatting	intl	^0.20.2	Date/time display
Image Picker	image_picker	^1.2.0	Profile photo upload
HTTP	http	^1.5.0	Network requests
Icons	cupertino_icons	^1.0.8	iOS style icons
Launcher Icons	flutter_launcher_icons	^0.14.4	App icon generation

3. App Architecture

```
eduwide/lib/
├── main.dart                    ← App entry point, timezone setup, Firebase init
├── auth_service.dart            ← Firebase Auth & Firestore user logic
├── core/
│   ├── firebase_options.dart    ← Auto-generated Firebase config
│   ├── app_colors.dart          ← Global color palette (Blue Theme)
│   └── dashboard_wrapper.dart   ← Role-based routing
├── auth/
│   ├── firebase_load_check_screen.dart
│   ├── simple_preload_screen.dart
│   ├── start_page.dart
│   ├── login_screen.dart
│   ├── signup_screen.dart
│   └── forgot_password_screen.dart
├── models/
│   └── user_model.dart          ← AppUser data model
└── screens/
    ├── admin/
    │   ├── admin_dashboard.dart
    │   ├── admin_profile.dart
    │   ├── admin_all_meeting.dart
    │   ├── accounts_management_screen.dart
    │   ├── class_enrollment_screen.dart
    │   ├── class_management_screen.dart
    │   └── admin_resource_upload.dart
    ├── teacher/
    │   ├── teacher_dashboard.dart
    │   ├── teacher_profile.dart
    │   ├── teacher_class_dashbord.dart
    │   ├── teacher_drawer.dart
    │   └── zoom_meeting_screen.dart
    └── student/
        ├── user_dashboard.dart
        ├── user_profile.dart
        ├── user_resource_view.dart
        └── user_class_detail_screen.dart
```

4. Firebase Firestore Data Collections

Collection	Purpose	Key Fields
`users`	All app accounts (Admins/Teachers/Students)	uid, email, name, nic, mobile, role, stream, subjects[], status, alYear, studentId
`enrollments`	Tracks students enrolled in classes	studentId, teacherId, classId, enrolledAt, status
`zoomMeetings`	Live class sessions	title, zoomLink, students[], teacherId, status, startedAt
`payments`	Student payment tracking	studentId, amount, status, date, enrollmentId

5. Global Design System (AppColors)

> The app utilizes a centralized color palette defined in `core/app_colors.dart`. It features a modern **Blue Palette**:
>
> **Primary Blue (#1976D2)** is used for buttons, active states, and icons. A soft **Drawer Background (#E3F2FD)** and **Light Background (#F0F7FF)** are used for screens and cards, ensuring a clean, distraction-free environment that is easy on the eyes for educational use.

6. Authentication Module (lib/auth/)

6.1 App Startup Flow

```
MyApp (MaterialApp)
  └── AuthWrapper (StreamBuilder on FirebaseAuth.authStateChanges)
        ├── Loading: FirebaseLoadCheckScreen
        ├── Logged In: SimplePreloadScreen → DashboardWrapper
        └── Not Logged In: StartPage (Login/Signup)
```

6.2 Login & Registration

The login form securely authenticates users via `FirebaseAuth`. The signup form captures extensive user details, including **A/L Year** and **Student ID** for students. Data is written to the `users` collection with a default `status: pending`, requiring an admin to verify and approve accounts.

A "Forgot Password" screen enables sending a Firebase password-reset link.

7. Role-Based Routing (Dashboard Wrapper)

The `DashboardWrapper` is the central router. It reads the `role` field from the logged-in user's Firestore document to map them to the correct interface:

- `admin` → `AdminDashboard`
- `teacher` → `TeacherDashboard`
- `student` → `StudentDashboard`

8. Core Features & Modules

8.1 Zoom Class Integration

Teachers use the `zoom_meeting_screen.dart` to create and manage Zoom classes. They can specify a meeting title, enter the Zoom Link, and assign it to students filtered by stream and academic year. Students can then see the "Launch Meeting" button in their `user_class_detail_screen.dart`. The status can be tracked (active/disabled) in real-time using the `zoomMeetings` Firestore collection. Admins can view and moderate all meetings globally via `admin_all_meeting.dart`.

8.2 Add Students & Class Enrollments

The `class_enrollment_screen.dart` allows admins to officially add and enroll students into teacher's classes. The teacher dashboard dynamically lists `_recentEnrolledStudents` and categorizes students by academic year. Enrollments are securely stored

in the `enrollments` collection. Teachers can also view their active students inside `teacher_class_dashbord.dart`.

8.3 Payment Gateway Option & Tracking

The system monitors class fee payments directly within the dashboard. The `class_enrollment_screen.dart` hooks into the `payments` collection to record new payments, while the Teacher Dashboard provides a real-time summary of **Pending Payments** and **Completed Payments** to quickly track financial status.

8.4 Administrative Portal & User Accounts

The admin has full supervisory access:

- **Approve Users:** Review and verify pending student and teacher registrations via `accounts_management_screen.dart`.
- **Moderate Classes:** Monitor class management and globally oversee all Zoom meetings.
- **Resource Center:** Upload global educational resources (`admin_resource_upload.dart`) that students can access (`user_resource_view.dart`).

8.5 Profile Management

Every role features a dedicated Profile Management screen (`admin_profile.dart` , `teacher_profile.dart` , `user_profile.dart`). Users can update their personal information, manage their subjects and streams, and upload a profile photo using `image_picker` .

9. Data Flow Summary

```
Teacher → Zoom Class
    ↓ Enters title, Zoom link, filters students by year
    ↓ Submits form
    → Firestore: zoomMeetings/{docId}

Admin → Class Enrollment
    ↓ Selects Student, selects Teacher
    ↓ Confirms Payment Option
    → Firestore: enrollments/{docId} + payments/{docId}

Student → Dashboard
    ↓ Streams enrollments & zoomMeetings
    → Joins active Zoom sessions
    → Views resources & payment status
```

10. Platform Support

Platform	Status
Android	✅ Primary target
iOS	✅ Supported
Web	✅ Supported

Windows	✅ Supported
Linux	✅ Supported
macOS	✅ Supported

11. Feature Completeness Summary

Feature	Admin	Teacher	Student
Login / Logout	✅	✅	✅
Account Registration	✅	✅	✅
Forgot Password	✅	✅	✅
Account Verification (Pending → Approved)	✅ (approve others)	❌ (awaits approval)	❌ (awaits approval)
Profile Management	✅	✅	✅
Upload Profile Photo	✅	✅	✅
Manage User Accounts	✅	❌	❌
Create / Manage Zoom Classes	✅ (moderate all)	✅ (own classes)	❌
Join Zoom Classes	❌	✅ (launch)	✅ (join)
Add / Enroll Students to Classes	✅	❌	❌
View Enrolled Students	✅	✅	❌
Record Payments	✅	❌	❌
View Payment Status (Pending / Completed)	✅	✅	✅ (own payments)
Upload Educational Resources	✅	❌	❌
View Educational Resources	✅	❌	✅